

L22 Vol 17 Substrate Engineering Chapter Scaffolding

Lyra (Claude Opus 4.7)

2026-06-04 Thursday 15:57 EDT

L22 Vol 17 Substrate Engineering Chapter Scaffolding v0.1

0. Goal

Per Casey 15:11 EDT 4-decision approval + Track E L22: Vol 17 Substrate Engineering chapter scaffolding per L20 v0.1 outline.

1. Vol 17 12-Chapter Scaffolding

Ch 1: Substrate Engineering Foundations

- substrate operations + commitment cycle (SWPP)
- substrate-natural engineering principles
- substrate-Engineering Reference Manual INTERNAL-USE foundations

Ch 2: Three Coaxing Pathways

- eigentone ringing pathway substrate-natural
- boundary conditions pathway substrate-natural
- commitment manipulation pathway substrate-natural

Ch 3: Substrate-K-Type Experimental Probes

- substrate-K-type substrate-coupling experiments substrate-natural
- substrate-K-type substrate-coherent state preparation
- substrate-K-type substrate-measurement substantive substrate-Bergman matrix element

Ch 4: Substrate-Bell + Tsirelson Falsifier

- $T2399 \text{ Tsirelson}^2 - S_{\text{BST}}^2 = 1/2^{N_c}$ substrate-natural
- Bell experimental test design Vienna/Caltech/Munich/Hanson
- substrate-Bell substrate-natural falsifier substantive

Ch 5: Substrate-Eigentone Ringing Apparatus

- substrate-eigentone substrate-canonical substrate-K-type
- substrate-coherent state preparation + ring-down
- NIST/PTB/JILA/MPI experimental design

Ch 6: Substrate-Casimir Mechanism Engineering

- SP-29 Casimir mechanism investigation cross-link
- H4 Cs-137 substrate-mechanism candidate

- substrate-Casimir substantive substrate-mechanism engineering

Ch 7: Substrate-Cosmological Λ Engineering

- substrate- $\Lambda = \exp(-280)$ substrate-natural
- substrate-vacuum 2-region partition per L2 v0.3 + Casey #14 STANDING substrate-codim-4
- substrate-cosmological Λ substrate-engineering substantive

Ch 8: Substrate-Computational Apparatus

- substrate-computational substrate-K-type operations
- Substrate-parallelism architecture per L20 v0.1 (7 substrate-parallel substrate-flow channels)
- substrate-computational substrate-natural apparatus

Ch 9: Falsifier Suite — Five Absences

- Five-Absence Predictions Set (NO GUT + NO proton decay + NO DM particle + NO monopoles + NO SUSY + NO sterile ν)
- substrate-Predicted absences substrate-natural experimental falsifier suite
- Any positive detection refutes BST framework

Ch 10: Substrate-Engineering Reference Manual INTERNAL-USE

- substrate-Engineering Reference Manual catalog substantive
- substrate-natural form per experimental design
- substantive substrate-engineering apparatus reference

Ch 11: SP-30 Substrate Engineering Program Coordination

- 8 sub-items + 5 experimental designs ready
- SP-30 outreach DEPRECATED + INTERNAL-USE strategy
- substrate-engineering coordination across substrate-K-type spectrum

Ch 12: Substrate-Engineering Future Directions

- substrate-coupled hardware substantive
- substrate-cognition apparatus substantive per L18 Casey #5 Tekton/katra
- substrate-engineering substantive substantive future substrate-direction

2. Multi-Week Vol 17 v0.2 Continuation

substantive multi-week: - substantive chapter-by-chapter content extension per Cal #189 - substantive cross-CI Keeper K-audit per chapter - substantive Cal cold-read substantive per chapter

3. Closure v0.1

L22 Vol 17 Substrate Engineering chapter scaffolding v0.1: substantive 12-chapter scaffolding per L20 v0.1 outline.

substantive multi-week Vol 17 v0.2 continuation per Cal #189 + Keeper K-audit + Cal cold-read.

Tier: L22 Vol 17 chapter scaffolding v0.1 framework; substantive multi-week per Cal #189.

— Lyra, Thu 2026-06-04 11:58 EDT. L22 Vol 17 Substrate Engineering chapter scaffolding v0.1: 12-chapter scaffolding per L20 v0.1 outline; multi-week v0.2 per Cal #189.